Real and Complex Unconditional Constants Are Different for a Five-Character Basis

Candidate full affirmative answer to the final question in arXiv:0703236

13 August 2026

Status. Candidate full result, subject to human review. For the explicit frequency set $\Lambda = \{0, 1, 2, 3, 5\}$, a static interval certificate proves

$$K_{\mathbb{C}}(\Lambda) > \frac{49999}{26400} > 1.89 > 1.888 > K_{\mathbb{R}}(\Lambda).$$

Thus the real and complex unconditional constants of a finite character basis can differ.

1 Question and current-status evidence

For a finite $\Lambda \subset \mathbb{Z}$, let

$$C_\Lambda = \operatorname{span}\{e_\lambda : \lambda \in \Lambda\} \subset C(\mathbb{T}), \qquad e_\lambda(z) = z^\lambda,$$

with the supremum norm. Write $M_u(\sum c_\lambda e_\lambda) = \sum u_\lambda c_\lambda e_\lambda$. The real and complex unconditional constants are

$$K_{\mathbb{R}}(\Lambda) = \max_{u_\lambda \in \{-1,1\}} \|M_u\|, \qquad K_{\mathbb{C}}(\Lambda) = \max_{|u_\lambda|=1} \|M_u\|.$$

Neuwirth's source paper proves equality for every three-character basis and ends by asking whether the constants can be different for larger finite sets [1]. The exact source passage is reproduced below.

arbitrarily large albeit finite, may in fact be proved by the method of Riesz products in [8, Appendix V, §1.II]; see also [14, Proposition 13.1.3]. The case of infinite sets remains open.

A second motivation was to show that the real and complex unconditional constants of the basis $(\mathrm{e}_{\lambda_1}, \mathrm{e}_{\lambda_2}, \mathrm{e}_{\lambda_3})$ of $\mathscr{C}_\Lambda$ are different; we prove however that they coincide, and it remains an open question whether they may be different for larger sets. The *real unconditional constant* of $(\mathrm{e}_{\lambda_1}, \mathrm{e}_{\lambda_2}, \mathrm{e}_{\lambda_3})$ is the maximum of the norm of the eight unimodular relative Fourier multipliers (t_1, t_2, t_3) such that $t_k = 0$ modulo π. Let i, j, k be a permutation of $1, 2, 3$ such that the power of 2 in $\lambda_i - \lambda_k$ and in $\lambda_j - \lambda_k$ are equal. Lemma 2.1 shows that the four relative multipliers satisfying $t_i = t_j$ modulo 2π are isometries and that the norm of any of the four others, satisfying $t_i \neq t_j$ modulo 2π, gives the real unconditional constant. In general, the complex unconditional constant is bounded by $\pi/2$ times the real unconditional constant, as proved in [18]; in our case, they are equal.

Figure 1: The open question in arXiv:0703236, PDF page 23.

A later posting by the same author computes the real constant for $\{0,1,2,3\}$ but explicitly says the general distinction remains undecided [2]. Its manuscript is dated 2013 and it was posted to arXiv in 2026.

have their modulus bounded by 3/5 for each τ, so that $5/3 \le S(\{0,1,2,3\}) < \sqrt{3}$.

A motivation for this problem is that we wish to know whether the real and complex unconditionality constants are distinct for basic sequences of characters z^n, but this remains undecided.

Figure 2: Later status evidence, arXiv:2603.28229, PDF page 1.

2 The result

Theorem 1. *For* $\Lambda = \{0,1,2,3,5\}$,

$$K_{\mathbb{C}}(\Lambda) > \frac{49999}{26400} > 1.89 \quad \textit{and} \quad K_{\mathbb{R}}(\Lambda) < 1.888 = \frac{236}{125}.$$

In particular,

$$K_{\mathbb{C}}(\Lambda) - K_{\mathbb{R}}(\Lambda) > \frac{779}{132000}.$$

The proof has an elementary reduction and two finite certificates. All certificate data and the outward-rounded checker are in `code/verify_gap.py`.

3 Two norm reductions

Lemma 2. *For every finite* Λ,

$$K_{\mathbb{C}}(\Lambda) = \sup_{0 \neq p = \sum c_\lambda z^\lambda} \frac{\sum_{\lambda\in\Lambda} |c_\lambda|}{\|p\|_\infty},$$
$$K_{\mathbb{R}}(\Lambda) = \max_{\varepsilon \in \{-1,1\}^\Lambda} \|L_\varepsilon\|, \qquad L_\varepsilon\Big(\sum c_\lambda z^\lambda\Big) = \sum \varepsilon_\lambda c_\lambda.$$

Proof. For the first identity, every complex multiplier satisfies $|\sum u_\lambda c_\lambda z^\lambda| \le \sum |c_\lambda|$. Conversely, for fixed coefficients choose $u_\lambda = \overline{c_\lambda}/|c_\lambda|$ (arbitrarily if $c_\lambda = 0$) and evaluate at $z = 1$.

For the second identity, fix a sign pattern ε. If $|\zeta| = 1$ and $d_\lambda = c_\lambda \zeta^\lambda$, then

$$\Big\|\sum d_\lambda z^\lambda\Big\|_\infty = \Big\|\sum c_\lambda z^\lambda\Big\|_\infty, \qquad \sum \varepsilon_\lambda d_\lambda = (M_\varepsilon p)(\zeta).$$

Taking the two suprema gives $\|M_\varepsilon\| = \|L_\varepsilon\|$. □

4 The complex lower certificate

Consider

$$p(z) = \frac{1}{100000}\big(19563 + 18666z + (-6125 - 28872i)z^2 + (18692 + 10604i)z^3 + (-5823 + 9055i)z^5\big).$$

The three nonreal coefficient moduli have the exact integer brackets

$$\begin{aligned} 29514^2 &\le 871108009 \le 29515^2, \\ 21490^2 &\le 461835680 \le 21491^2, \\ 10765^2 &\le 115900354 \le 10766^2. \end{aligned}$$

Consequently

$$\sum_{\lambda\in\Lambda} |c_\lambda| \ge \frac{19563 + 18666 + 29514 + 21490 + 10765}{100000} = \frac{99998}{100000},$$

whereas the corresponding upper brackets give

$$\|p'\|_\infty \le \sum \lambda |c_\lambda| < \frac{49}{25}.$$

At each of the 4096th roots of unity, outward-rounded interval evaluation gives

$$\max_{0\le k<4096} |p(e^{2\pi i k/4096})| < 0.52631.$$

Every point of $\mathbb{T}$ lies at angular distance at most $\pi/4096$ from this grid. The mean-value estimate and $\pi < 22/7$ therefore yield

$$\|p\|_\infty < \frac{52631}{100000} + \frac{49}{25}\frac{22}{7\cdot 4096} = \frac{3378009}{6400000} < \frac{66}{125}.$$

Lemma 2 now proves

$$K_{\mathbb{C}}(\Lambda) > \frac{99998/100000}{66/125} = \frac{49999}{26400}.$$

5 The real upper certificate

We use the following standard dual certificate. If a finite complex measure μ on $\mathbb{T}$ satisfies

$$\int_{\mathbb{T}} z^\lambda \, d\mu(z) = \varepsilon_\lambda \quad (\lambda \in \Lambda),$$

then $L_\varepsilon(f) = \int f \, d\mu$ and hence $\|L_\varepsilon\| \le \|\mu\|_{\mathrm{TV}}$.

It suffices to take $\varepsilon_0 = 1$, since simultaneous sign reversal does not change the norm. For each of the resulting 16 patterns, the checker stores a measure ν whose atoms have the exact form

$$\frac{a}{10^{12}} e^{-2\pi i p/120} \, \delta_{e^{2\pi i t/120}}, \qquad a, p, t \in \mathbb{Z}, \ a \ge 0.$$

Thus $\|\nu\|_{\mathrm{TV}} = \sum a/10^{12}$ exactly. Its five moment residuals are evaluated by outward-rounded interval arithmetic.

For completeness, the rounding residual is repaired analytically rather than ignored. Put

$$q_\lambda = \varepsilon_\lambda - \int z^\lambda \, d\nu(z) \quad (\lambda \in \Lambda), \qquad q_4 = 0,$$

let $\omega = e^{2\pi i/6}$, and define

$$b_k = \frac{1}{6}\sum_{j=0}^{5} q_j \omega^{-kj}, \qquad \eta = \sum_{k=0}^{5} b_k \delta_{\omega^k}.$$

Discrete Fourier inversion gives $\int z^j \, d\eta(z) = q_j$ for $0 \le j \le 5$, while

$$\|\eta\|_{\mathrm{TV}} \le \sum_{j=0}^{5} |q_j| = \sum_{\lambda\in\Lambda} |q_\lambda|.$$

Therefore $\mu = \nu + \eta$ is an exact representing measure and

$$\|L_\varepsilon\| \le \|\nu\|_{\mathrm{TV}} + \sum_{\lambda\in\Lambda} |q_\lambda|.$$

The following are decimal displays of rigorous interval upper bounds from the static checker. Each has been rounded upward to the shown sixth decimal.

ε	bound	ε	bound
+----	1.872706	+---+	1.666667
+--+-	1.414214	+--++	1.751844
+-+--	1.000001	+-+-+	1.521691
+-++-	1.887888	+-+++	1.666667
++---	1.751844	++--+	1.414214
++-+-	1.666667	++-++	1.872706
+++--	1.666667	+++-+	1.887888
++++-	1.521691	+++++	1.000001

In particular, all 16 exact corrected measures have total variation less than 1.888, so Lemma 2 gives $K_{\mathbb{R}}(\Lambda) < 1.888$.

Proof of Theorem 1. The complex and real estimates were proved in the preceding two sections. Finally,

$$K_{\mathbb{C}}(\Lambda) - K_{\mathbb{R}}(\Lambda) > \frac{49999}{26400} - \frac{236}{125} = \frac{779}{132000} > 0.$$

□

6 Verification and novelty assessment

The checker contains no optimization call and no downloaded data: it consists only of the displayed rational polynomial, the finite atom lists, exact integer/rational arithmetic, and outward-rounded evaluations of roots of unity using `mpmath.iv`. It checks the coefficient brackets, the 4096 grid, the derivative correction, all 16 normalized sign patterns, the DFT residual bound, and the final rational gap. A clean run prints

```
complex witness grid upper: 0.526309881899435
certified complex constant lower bound: 1.893901515151515
worst normalized real sign: +-++-
worst corrected real certificate: 1.887887419718339
CERTIFIED: K_C > 49999/26400 > 1.89 > 1.888 > K_R
certified gap lower bound: 779/132000
```

The local run indexes and a bounded arXiv/web search on 13 August 2026 found no prior example separating the two constants. The later same-author status paper still calls the distinction undecided. The construction and constants above should therefore be treated as a new candidate full answer, with high correctness confidence from the static certificate and moderate-to-high novelty confidence pending specialist review.

References

[1] S. Neuwirth, *The maximum modulus of a trigonometric trinomial*, arXiv:math/0703236 (2007; revised 2008).

[2] S. Neuwirth, *On the (Fourier analytic) Sidon constant of* $\{0, 1, 2, 3\}$, arXiv:2603.28229 (posted 2026; manuscript dated 2013).